



# Enhancing Fraud Detection in Credit Card Transactions: A Comparative Study of Machine Learning Models

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## Abstract

The convenience and efficiency of credit cards for online transactions have led to their widespread adoption. However, this increased usage has also heightened the risk of credit card misuse and fraud. Such fraudulent activities result in substantial financial losses for both cardholders and financial institutions. The primary objective of this research study is to detect fraudulent activities using machine learning techniques, which offer promising avenues for accurate identification. However, the challenge lies in developing robust models capable of effectively distinguishing between genuine and fraudulent transactions. This study addresses this gap by proposing a systematic methodology for fraudulent transaction detection using machine learning (ML) algorithms. Three datasets are used, including two balanced datasets and one unbalanced dataset, as fraudulent transactions are typically far less frequent than genuine ones, ensuring the model's generalizability. Through preprocessing techniques such as handling missing values and reducing dimensionality, combined with the implementation of machine learning models (Support Vector Machine (SVM), Decision Tree (DT)), ensemble learning (EL) models (Random Forest (RF)), Extreme Gradient Boosting (XGBoost), and Adaptive Boosting (AdaBoost), and deep learning models (Multilayer Perceptron (MLP), Artificial Neural Network (ANN)), the study explores various models' effectiveness in capturing complex relationships within transactional data. The results demonstrate significant success, DT and RF, and MLP models achieving high accuracy of 0.99 underscoring their potential for accurate fraud detection in financial transactions. In the case of the unbalanced dataset, the RF model also performed effectively, achieving an accuracy of 0.97, with improved precision and recall for fraud detection.

**Keywords** Fraud · Credit card · Machine learning · Financial

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## 1 Introduction

Since 1996, the background of banking service has changed a lot, following the use of internet banking by Citibank and Wells Fargo Bank from the US. This innovation paved the way for widespread adoption of online transactions, including the use of credit cards for various purposes (Madan et al., 2022). During the last decade the rise of e-commerce, online payment systems, telecommuting, digital banking and social networking platforms has contributed even more to online payments (Carcillo et al., 2018). On the other hand, this rise in internet transaction made the online payment system also more attractive to fraudster targeting and preying on vulnerabilities in different online payment systems (Maniraj et al., 2019; Maesa & Mori, 2020). As digital technologies continue to advance, traditional cash transactions are gradually being replaced by digital platforms, reshaping the landscape of financial management. The integration of technology into digital transactions has become essential for maintaining productivity and gaining a competitive edge in the market (Pencarelli, 2020). As a result, internet-based banking and other digital transaction methods—which mostly involve the usage of credit cards—have become popular choices for consumers who want to do financial transactions conveniently from the convenience of their homes or places of employment. A credit card, as described by experts (Benson Edwin Raj & Annie Portia, 2011), is a plastic card embedded with personal information issued by financial institutions, enabling users to make purchases globally at their convenience. However, the misuse of credit cards, whether physically or digitally, for unauthorized transactions constitutes credit card fraud, often resulting in significant financial losses (Carcillo et al., 2021). Today, the digital environment has made it easier for fraudsters to perpetrate such crimes, as transactions can be completed without the physical presence of the card, relying solely on its information (Lakshmi & Kavila, 2018). Besides that, the development of credit card systems has impacted in numerous directions, and both huge enterprises and little firms have had to adapt their monetary strategy and commercial strategies.

According to the recent global statistics, there has been a continual upsurge in the cases of credit card fraud on the yearly basis, with the rate of 46% rise reported the following year, according to the recent global statistics. Because the trading volume of credit cards is now more active, the frequency of the attempts to deceive also increases. The case was also paralleled by the rise of credit card fraud on online sellers, especially e-commerce merchants, where the numbers went up by 140% in the past three years. Out of all countries, the U.S. is the epi-center of digital threat with 46% of all credit card fraud loss that happen outside the American borders. The financial loss caused by credit card fraud is a major issue, and researchers expect a global damage of no less than \$43 billion by the year 2026. The United States is, in fact, one of the major and extremely attractive markets for fraudsters. Losses due to fraud with Credit card are expected to reach \$12.5 billion over the next two years. They proclaim the dire necessity of the adoption of tough anti-fraud measures and bring advanced technologies to prevent innocent consumers and business entities to be exploited. As a result for this reason financial institutions should pay attention first for these automated fraud recognition systems. Supervised credit card fraud detection aims to develop ML models based on transactional credit card data to differenti-

ate between transactions that are fraud and not fraud and make decisions regarding incoming transactions (Alarfaj et al., 2022). This task presents challenges such as ensuring rapid system response, cost sensitivity, and feature preprocessing. ML, a field of Artificial Intelligence (AI), utilizes computer algorithms to make predictions built on past data trends (Thennakoon et al., 2019).

Recently, a variety of ML approaches have been utilized to detect frauds with credit card such as logistic regression, decision tree, support vector machine and ensemble methods (Mytnyk et al., 2023). While these models have demonstrated moderate success, they are often limited by their high false positive rates and poor adaptability to evolving fraud patterns. Furthermore, deep learning architectures like convolutional and recurrent neural networks are also being applied by several studies but most of them either utilize oversimplified datasets or do not consider real world class imbalance problems (Nguyen et al., 2020). It is limited by these limitations and cannot be practically implemented in the dynamic financial environments. There is, therefore, a wide gap to be closed in terms of developing such models that are accurate yet generalizable, interpretable, and instantly applicable (i.e., real time).

The latest generation of deep learning (DL) algorithms are being employed in a range of domains such as computer networks, intrusion detection (Mahindru & Sangal, 2022), banking, insurance (Sharma & Harithas, 2019), and malware detection (Qiu et al., 2021). However, there is very little study on the application of deep neural networks specifically for identifying credit card fraud.

This study aims to develop and test ML algorithms for fraudulent transaction identifications in the credit cards in order to reduce the losses to an individual card holder or the banks. The main contributions of the study are as follows:

- Development of robust ML models for exact detection of fraudulent activities in credit card transactions.
- Exploration of preprocessing techniques, feature selection methods, and model evaluation strategies to optimize the fraud detection systems results.

This study addresses critical gaps in the current literature on credit card fraud detection, where existing models often suffer from high false positive rates and limited adaptability to diverse transactional data. This research made contributions in terms of developing robust ML models, RF, DT and Multilayer Perceptron optimized to detect the fraud most accurately, utilizing processed data using comprehensive preprocessing such as handling missing values, multicollinearity and dimensionality reduction. Furthermore, we consider feature selection and model interpretability which increase detection accuracy and reduce number of false alarms by evaluating the models using, for example precision, recall and F1 score.

## 2 Literature Review

Scholars have used different techniques covering the spectra of ML, EL and DL in identifying transactions that are termed fraud. These data mining strategies include the logistic regression approach, DT technique, and SVM, which have been effec-

tively used to identify fraudulent patterns within credit card transactions (Afriyie et al., 2023; Dhankhad et al., 2018; Maniraj et al., 2019). In addition, EL methods, including random forests and gradient boosting machines have received attention because they use multiple base learners to enhance the prediction accuracy (Do et al., 2023). In recent years, the advent of DL algorithms, including convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, has added new avenues for detecting fraudulent behavior by leveraging the capacity of deep hierarchical representations (He, 2022). These studies were selected for review based on their relevance to widely adopted and high-performing techniques in each learning category, forming a representative foundation for benchmarking our selected models.

## 2.1 Machine Learning

Artificial neural networks (ANN) have proved to be essential instruments for dealing with a large number of difficult problems: fraudulent activity in banking transactions to global trends online. Thanks to these algorithms, it is possible to quickly and accurately find where fraudulent activity needs to be alerted by automatically learning from data trends. Past transaction data can be analysed in order to extract relevant elements which can be analysed by ML algorithms in order to detect unusual patterns of transactions suggestive of fraud. In a comprehensive approach involving multiple machines learning models, Thennakoon et al., (2019) suggest real time detection and handle four most important fraud categories. While this helps to choose best prediction algorithms, we also acknowledge that further development in prediction ranges and destiny extensions to tackle place based frauds. Moreover, Jain et al. (2019) conduct a relative study on a range of ML models, emphasizing the importance of accurate and preemptive fraud detection systems. Their evaluation based totally on quantitative measurements consisting of accuracy and detection charge underscores the need to cope with false alarm charges and adapt to evolving fraud processes for more suitable effectiveness.

Several studies delve deeper into precise ML methods for credit card fraud detection (Ryman-Tubb, Krause, & Garn, 2018; Sharma & Panigrahi, 2013). Lakshmi and Kavila, (2018) discover the performance of logistic regression (LR), DT, and RF techniques on rather imbalanced datasets, highlighting the significance of oversampling to balance the statistics. Moreover, RB & KR, (2021) suggest the usage of a couple of ML algorithms alongside a DL approach, reaching near-ideal accuracy with the ANN. Additionally, Maniraj et al., (2019) search on anomaly detection algorithms for credit score card fraud detection, achieved high levels of accuracy but problems with precision due to skewed dataset. Lastly, Afriyie et al. (2023) concentrate on logistic regression, DT, and RF algorithms, concludes that RF is the best-fit model but admits possible drawbacks, such as the failure to consider other approaches and applicability of the results to other areas or industries. The overview of ML centered Techniques is presented in Table 1.

**Table 1** Summary of present ML techniques

| Ref                       | Year | Method(s)                                                    | Dataset                                                   | Size                              | Limitation                                                          | Result            |
|---------------------------|------|--------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------|-------------------|
| (Thennakoon et al., 2019) | 2019 | LR, NB, LR and SVM                                           | fraud transactions log file and all transactions log file | 917,781 records                   | Average prediction accuracy                                         | Acc of SVM: 91%   |
| (Jain et al., 2019)       | 2019 | SVM, ANN, Bayesian Network, KNN, Fuzzy Logic, Decision Trees | KDD                                                       | 125,973 records                   | False alarm rates and adaptability to evolving fraud tactics        | Acc of ANN: 99.71 |
| (Lakshmi & Kavila, 2018)  | 2018 | LR, DT, RF                                                   | European bank credit card transactions                    | 2,84,808 credit card transactions | No use of time series based algorithms                              | Acc of RF: 95.5   |
| (RB & KR, 2021)           | 2021 | SVM, KNN, ANN                                                | credit card transactions                                  | 492 transactions                  | reliance on preprocessing techniques to address dataset imbalance.  | Acc of ANN: 100   |
| (Maniraj et al., 2019)    | 2019 | Local Outlier Factor, Isolation Forest Algorithm             | credit card transactions                                  | 25,000 transactions               | Low precision, especially with smaller subsets of the dataset,      | Acc: 99.6         |
| (Afriyie et al., 2023)    | 2023 | LR, DT, RF                                                   | credit card transactions                                  | 96,961 transactions               | potential lack of generalizability to different regions or sectors. | Acc of RF: 96     |

## 2.2 Ensemble Learning

Ensemble mastering algorithms have been identified as useful approach towards enhancing the predictive power of models by employing a number of base learners. Ensemble methods again take advantages of the heterogeneity between learners; concept of combining multiple models make them superior in terms of performance as well as their ability to generalize. These algorithms provide a generalized approach in order to solve complex problems in multiple areas of interest including, fraudulent transaction identification in financial networks. Bagging, boosting and stacking method involves the combined utilization of multiple models and gives a better security approach to apparent fraud as identified by various models. Like the one, Dhankhad et al. (2018) explain the use of supervised ML algorithms, trying to increase the detection accuracy through the EL method. Instead, their work is aimed at selecting relevant predictors, as well as evaluating the performance of the algorithms against the super classifier developed by the authors. However, some challenges arise concerning handling the dataset imbalance because data for this study and analysis gathered from the publicly available dataset may limit the generalization of the findings to possible future occurrences. Similarly, Do et al. (2023) propose a new approach for early detection of fraudulent credit card transactions with the aid of combining behavioral evaluation strategies with the XGBoost supervised ML algorithms. They introduce new features for behavioral analysis and demonstrate the effectiveness of their approach in improving detection efficiency, particularly with imbalanced datasets. However, similarly research is warranted to deal with

anomaly detection in unbalanced datasets and reduce fake alarms concerning normal transactions.

Alternatively, Xuan et al., (2018) explore the efficacy of RF trained on historical transaction data to detect credit card fraud, utilizing ML techniques. Their study compares the performance of RF with base classifiers and analyzes their effectiveness in fraud detection by using records from a Chinese e-commerce organisation. However, the study's focus on RF algorithms may overlook other EL techniques that might probably enhance fraud detection accuracy. Also, Fang et al. (2019) introduce the usage of the Smote algorithm to deal with statistics imbalance in credit card fraud detection, with a focal point on assessing the Light Gradient Boosting Machine (LightGBM) model. Comparing it with different algorithms, they exhibit LightGBM's superior overall performance in detecting credit score card fraudulence. However, the reliance on client transaction facts for evaluation may additionally restrict the generalizability of findings. The overview of EL based Techniques is shown in Table 2.

### 2.3 Deep Learning (DL)

DL algorithms are a subset of ML approaches that draw inspiration from the architecture and operation of neural networks seen in the human brain (Patidar & Sharma, 2011). These algorithms excel at learning complex patterns and representations immediately from statistics, making them specifically properly-acceptable for tasks related to huge datasets and intricate relationships. Taken together, DL models (CNNs, recurrent neural networks (RNNs) ... etc.) have been able to successfully perform in many domains such as; computer vision, natural language processing, and speech recognition. Based on the notion of fraudulent transaction detection, DL algorithms have powerful competencies for recognizing veiled patterns of fraudulent behavior in financial transaction info. For example, in order to help choose the best method for detecting credit card fraud, Sadgali et al., (2019) compares various machine learning approaches with an emphasis on neural networks. Their assessment of data mining strategies including NN, MPL, and CNN, dataset highlights the MLP as the best-performing technique, with the intention of integrating it into an adaptive fraud detection model. However, the study's limitation lies in its contracted focus on best three par-

**Table 2** Overview of existing EL techniques

| Ref                     | Year | Method(s)                              | Dataset                                        | Size                               | Limitation                                       | Result              |
|-------------------------|------|----------------------------------------|------------------------------------------------|------------------------------------|--------------------------------------------------|---------------------|
| (Dhankhad et al., 2018) | 2018 | Ensemble Learning and Machine Learning | Real-world dataset on credit card transactions | 284,807 transactions               | Limited generalizability                         | Acc of Stacking: 95 |
| (Xuan et al., 2018)     | 2018 | Random Forest                          | e-commerce company of China                    | 30,000,000 individual transactions | Focus on random forest algorithms only           | Acc: 96.77          |
| (Fang et al., 2019)     | 2019 | LGBM, GBM, RF                          | Client transaction data                        | 284,807 transactions               | No use of time series based algorithms           | Acc of LGBM: 0.99   |
| (Do et al., 2023)       | 2023 | XGBoost algorithm                      | VestaCorporation. Regarding Vesta              | 590,540 transactions               | High false alarms regarding normal transactions. | Acc: 99.0           |

ticular ML techniques, probably overlooking other techniques. Likewise, Karkaba et al. (2023) extend this effort to study the effectiveness of different DL topologies and parameters with an actual credit card fraud dataset with millions of transactions and distributed cloud computing. They undertake an analysis to provide understanding on sensitivity of the model parameters and provide a guideline on parameter tuning for the financial institutions' Losses reduction due to fraudulent activity. However, the study needs to be done to see how best the models perform depending on the network size and other hyperparameters that have not been included in the study.

Nguyen et al., (2020) explore the use of DL techniques in credit card fraud detection and discuss how these methodologies perform against classical ML methods on various datasets of a financial nature. It proves that DL algorithms such as CNN and LSTM generally show great results in improving the accuracy of fraud detection than using norms algorithms. However, the major limitation is possible negative changes in performance on unseen data due to experimental manipulations of various sampling methods for addressing class imbalance, which means that further investigation of hyperparameters for improving the results could be useful. Alarfaj et al. (2022) investigate the DL algorithms for credit card fraud detection and then analyses the performance of DL algorithms with respect to the European card benchmark dataset in comparison with the classical ML. They show how their approach gains better performance in detecting fraud against preset performance measurements. Still, the problem of how to use these approaches in practical setting and, in particular, to control false negative rates stays an issue.

Several recent studies have investigated the use of Multilayer Perceptron (MLP) models for credit card fraud detection due to their effectiveness in capturing complex patterns in transactional data. Bassam Kasasbeh et al. (Kasasbeh et al., 2022) proposed an MLP-based ANN model aimed at improving detection accuracy by evaluating multiple performance metrics. Fatima Zohra El Hlouli et al. (2020) compared MLP with Extreme Learning Machine (ELM) classifiers on a credit card fraud dataset to assess their performance and classification time. Aya Abd El Naby et al. (El Naby et al., 2021). integrated MLP into a deep learning framework alongside a CNN-based model (OSCNN) using the Kaggle dataset to enhance fraud prediction through oversampling techniques. The overview of Deep Learning based Techniques is shown in Table 3.

Existing studies (Tables 1, 2 and 3) on credit card fraud detection primarily focus on single models or limited ensemble techniques, emphasizing predictive accuracy but often facing high false alarm rates and limited adaptability to new fraud patterns (Jain et al., 2019). Techniques like RF are common (Xuan et al., 2018), yet they lack time series analysis, crucial for capturing evolving fraud tactics. Moreover, many studies, such as those by (Lakshmi & Kavila, 2018) and (Maniraj et al., 2019), use balanced datasets, which may not reflect real-world imbalanced data challenges.

This study addresses these gaps by employing a diverse set of models, including DT, RF, MLP, and SVM, with robust preprocessing and k-fold cross-validation for better generalization. Unlike prior work, we focus on adaptability to imbalanced datasets, making our approach more aligned with real-world fraud detection needs and enhancing the ability of system to handle evolving fraud patterns.

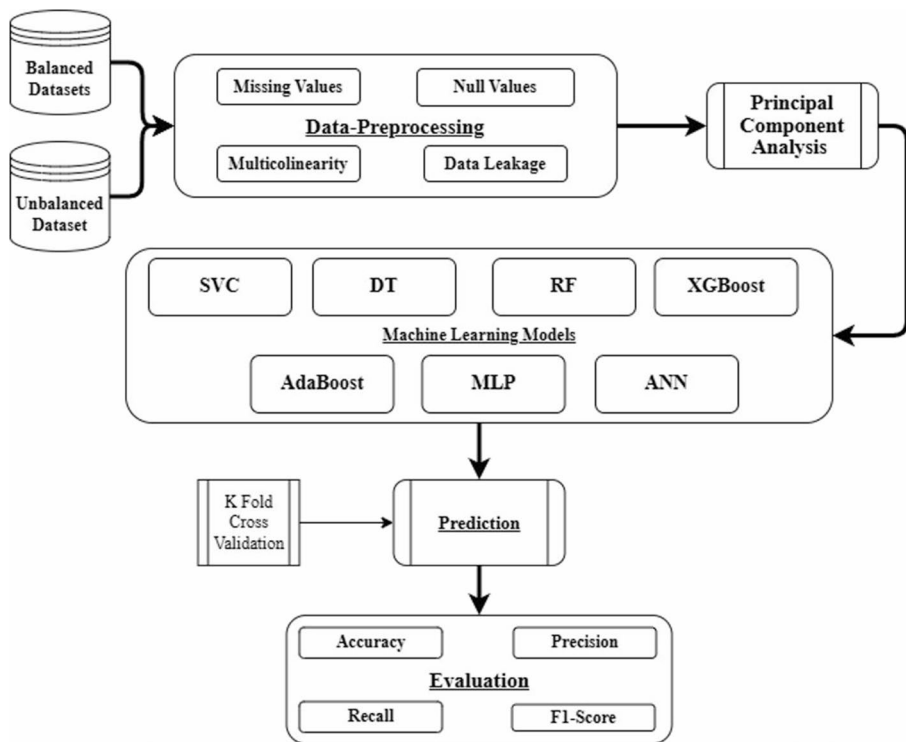
**Table 3** Overview existing DL techniques

| Ref                      | Year | Method(s)           | Dataset                                        | Size                                            | Limitation                                                                                              | Result                   |
|--------------------------|------|---------------------|------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------|
| (Sadgali et al., 2019)   | 2019 | Neural networks     | Bank Record                                    | 60,000 transactions                             | Focus on a limited set of machine learning techniques                                                   | Acc of MLP: 87.88        |
| (Karkaba et al., 2023)   | 2023 | ANN, LSTM, RNN, GRU | Real-world dataset on credit card transactions | 80 million pre-labeled credit card transactions | determine optimal model performances with varying network sizes and additional hyperparameters          | Acc of GRU: 91.6         |
| (Nguyen et al., 2020)    | 2020 | CNN and LSTM        | ECD, SCD                                       | 284,807, 3075 Transactions                      | Potential decrease in performance on unseen data when using sampling methods to address class imbalance | F1 Score of LSTM: 84.85% |
| (Alarfaj et al., 2022)   | 2022 | CNN                 | Credit card dataset                            | 284,807 transactions                            | Potential challenge in effectively implementing proposed approaches in real-world scenarios             | Acc: 99.72%.             |
| (Kasasbeh et al., 2022)  | 2022 | MLP (ANN-based)     | Credit card dataset                            | 284,807 transactions                            | Manual transaction review not feasible                                                                  | High accuracy claimed    |
| (El Hlouli et al., 2020) | 2020 | MLP, ELM            | Credit card dataset                            | 284,807 transactions                            | No mention of imbalance handling                                                                        | Acc: 95.46%              |
| (El Naby et al., 2021)   | 2021 | MLP, CNN (OSCNN)    | Credit card dataset                            | 284,807 transactions                            | Only compared two models; CNN focus                                                                     | F-Measure: 99.49%        |

### 3 Methodology

In this research we employed the systematic procedure to identify the fraudulent transactions using ML techniques on three datasets. Dataset 1 (Balanced) comprises 550,000 anonymized credit card transactions from European cardholders during 2023. Dataset 2 (Balanced) contains 688 entries, including customer IDs with credit cards. Dataset 3 (Unbalanced) includes 14,446 credit card transactions from the western United States. The process begins by preprocessing the datasets to handle missing values, address multicollinearity, and apply dimensionality reduction through Principal Component Analysis (PCA). Subsequently, we use the machine learning models (SVM, DT), ensemble learning (EL) models (RF, XGBoost, and Adaboost), and deep learning models (MLP, ANN), detailing their ensemble approaches and feature importance calculations using techniques like Gini Importance and SHAP values. Additionally, SVM and DT are explored for their effectiveness in seizing complex relationships within the transactional data, as shown in Fig. 1. Traditional metrics like accuracy, precision, recall, and F1 score are used to assess the performance of the model. 10-fold cross-validation is used to guarantee robustness and reduce the possibility of overfitting.





**Fig. 1** Proposed methodology

**Table 4** Features of dataset 1

| Feature Name | Description                                                                                  |
|--------------|----------------------------------------------------------------------------------------------|
| id           | Unique identifier for each transaction.                                                      |
| V1-V28       | Anonymized features representing various transaction attributes (e.g., time, location, etc.) |
| Amount       | The transaction amount.                                                                      |
| Class        | Fraud label (0: Non-fraudulent, 1: Fraudulent).                                              |

### 3.1 Dataset

In this research, three datasets are used to evaluate the efficiency of fraud detection algorithms. Each dataset is described in detail below.

#### 3.1.1 Dataset 1

Dataset 1 comprises more than 550,000 anonymized records of credit card transactions conducted by European cardholders during 2023 (Elgiriye withana, 2023). This dataset is developed to support the improvement of fraud detection models by identifying possible fraudulent transactions. The features of Dataset 1 are explained in Table 4.

Dataset 1 is balanced, with equal representation of 284,315 instances for fraudulent (1) and non-fraudulent (0) transactions. This balance was achieved through prior curation and sampling by the original providers and was used as-is in our analysis without further resampling.

### 3.1.2 Dataset 2

Dataset 2 includes 689 records containing customer IDs and associated credit card details (Semyari, 2023). This dataset provides features specific to individuals, enabling the identification of fraudulent behavior, although the original features and detailed background information remain confidential. The features of Dataset 2 are explained in Table 5.

Dataset 2 contains 689 records, with a near-balanced distribution of classes, enabling fair evaluation of fraud detection models without significant bias toward the majority class.

### 3.1.3 Dataset 3

Dataset 3 consists of 14,446 credit card transaction records from the western United States, including 12,601 normal and 1,845 fraudulent transactions (Choudhury, 2024). This dataset includes detailed information about each transaction, such as customer details, merchant and category of purchase, and fraud labels, offering a balanced and diverse data foundation for evaluating model performance. The features of Dataset 3 are explained in Table 6.:

Dataset 3 is imbalanced, consisting of 14,446 transactions, with 12,601 normal (0) and 1,845 fraudulent (1) cases. This unbalanced character is a reflection of distributions in the actual world, where a small percentage of transactions are fraudulent.

This study uses datasets that combine balanced and imbalanced scenarios that create a great environment to evaluate fraud detection models. These datasets combined can well assess the effectiveness of the model along with conventional metrics such as F1 score, recall, accuracy, precision. While Dataset 1 and Dataset 2 enable fair and unbiased metric reporting, Dataset 3 provides insights into the models' effectiveness in handling imbalanced data, offering a realistic perspective for practical applications.

## 3.2 Preprocessing

To improve data quality, we adopted a systematic preprocessing pipeline involving correlation analysis, dimensionality reduction, normalization, and data integrity checks, as shown in Table 5.

**Table 5** Features of dataset 2

| Feature Name | Description                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer_ID  | Unique identifier for each customer.                                                                                                             |
| A_1 to A_14  | Features representing various attributes of the customer such as transaction frequency, average transaction value, past transaction history etc. |
| Class        | Fraud label (0: Non-fraudulent, 1: Fraudulent).                                                                                                  |

**Table 6** Features of dataset 6

| Feature Name          | Description                                              |
|-----------------------|----------------------------------------------------------|
| trans_date_trans_time | Timestamp of the transaction.                            |
| merchant              | Merchant associated with the transaction.                |
| category              | Purchase category.                                       |
| amt                   | Transaction amount.                                      |
| city                  | City where the transaction occurred.                     |
| state                 | State where the transaction occurred.                    |
| lat                   | Latitude of the customer's location.                     |
| long                  | Longitude of the customer's location.                    |
| city_pop              | Population of the city where the transaction took place. |
| job                   | Job title or occupation of the customer.                 |
| dob                   | Customer's date of birth.                                |
| trans_num             | Unique transaction identifier.                           |
| merch_lat             | Latitude of the merchant's location.                     |
| merch_long            | Longitude of the merchant's location.                    |
| is_fraud              | Fraud label (0: Non-fraudulent, 1: Fraudulent).          |

### 3.2.1 Handling Multicollinearity

We investigated multicollinearity among features by calculating Pearson correlation coefficients. In Dataset 1, variables V2 and V4, and V10 and V12, exhibited high correlations ( $>0.6$ ). In Dataset 2, features A3 and A5 showed correlation above the 0.6 threshold. In Dataset 3, amt and city\_pop were moderately correlated. To address these redundancies, Principal Component Analysis (PCA) was applied to transform the correlated variables into uncorrelated components while retaining significant variance.

### 3.2.2 Dimensionality Reduction

For Dataset 1, PCA retains 95% of the variance, reducing the 29 original features to 22 components, balancing model complexity and accuracy. This is especially beneficial for the large dataset of 550,000 records. Dataset 2 has 14 anonymized features (A1–A14), and PCA retains 90% of the variance, reducing them to 10 components, improving computational speed and interpretability. Dataset 3, with 15 features and an imbalanced nature, uses PCA to retain 95% variance, reducing it to 12 components, which helps address imbalance and improve model performance. A summary of dimensionality reduction under different thresholds is shown in Table 5.

### 3.2.3 Data Normalization

Standardization was applied to normalize the feature ranges, particularly for distance-based algorithms like SVM. All numerical features were transformed to have zero mean and unit variance.

### 3.2.4 Handling Missing and Null Values

We first examined each dataset for missing or null values. Dataset 1 had no missing entries. Dataset 2 contained a few missing values in A7 and A11, which were imputed using the median of each respective feature. Dataset 3 had null values in the job and city fields; these were treated using mode imputation. No rows were dropped to preserve data integrity.

### 3.2.5 Prevention of Data Leakage

To prevent data leakage, we ensured that the training and test datasets remained strictly separated, preventing any information from the test set from influencing the model during training. We also conducted a thorough review of the features to identify and remove any that could be directly or indirectly related to the target variable, ensuring that the model's performance metrics are reflective of real-world prediction capabilities. Finally, we ensured that no future information was included in the training process, particularly for time-sensitive data (Table 7).

**Table 7** Data dimensions after PCA

| Dataset   | Variance | Number of Features |
|-----------|----------|--------------------|
| Dataset 1 | 90%      | 17                 |
|           | 91%      | 18                 |
|           | 92%      | 19                 |
|           | 93%      | 20                 |
|           | 94%      | 21                 |
|           | 95%      | 22                 |
|           | 96%      | 23                 |
|           | 97%      | 24                 |
|           | 98%      | 26                 |
| Dataset 2 | 99%      | 27                 |
|           | 90%      | 10                 |
|           | 91%      | 11                 |
|           | 92%      | 12                 |
|           | 93%      | 13                 |
|           | 94%      | 14                 |
|           | 95%      | 14                 |
| Dataset 3 | 90%      | 11                 |
|           | 91%      | 12                 |
|           | 92%      | 13                 |
|           | 93%      | 14                 |
|           | 94%      | 14                 |
|           | 95%      | 12                 |

### 3.3 Models

This study's primary goal is to develop precise models that can distinguish between authentic and fraudulent transactions using a variety of supervised and EL techniques. This means that there are various approaches to this important problem, ranging from RF to SVM. These models were chosen due to their high performance in previous research (Gyamfi & Abdulai, 2018; Hashemi et al., 2023; Kasasbeh et al., 2022), interpretability, relevance to the real-time detection requirements covered in the literature review, and their capacity to handle imbalanced data.

#### 3.3.1 Support Vector Machine

More specifically, SVMs are advanced type of supervised ML models that works both in classification and regression types of problems (Alarfaj et al., 2022). SVMs' goal is to determine a hyperplane that quality separates statistical points from one-of-a-kind kinds of classes in feature house at the identical time as pushing the margin among those instructions (Zainuddin et al., 2018). This This margin adjustment gives SVMs the flexibility to adapt to a wide range of choice problems and makes them resistant to models present in the records.

In this study, SVMs play a crucial role of identifying fraudulent transactions through presenting a pivotal option. Due to the nature and often high-dimensional features extracted from transaction attributes within this task, SVMs can display sensible skill ability in modeling fine-grained data relations. This capability allows for the detection of dispersed patterns linked to fraud. However, SVM s is well known for its capability to deal with linear as well as non-linear relationship in the obligations the boundary among fraudulent and non-fraudulent could also be blurred and is not strictly directly relation.

$$f(x) = \sum_{i=1}^N \alpha_i y_i K(x, x_i) + b \quad (1)$$

Where,  $f(x)$ : Symbolizes the decision function that yields the predicted output for the given input  $x$ , which, in this case, represents the transaction attributes used for detection.  $N$ : Represents the number of support vectors, which are the key data points contributing to the decision function in the SVM model.  $\alpha_i$ : Symbolizes the Lagrange multipliers, determined during the training phase, which indicate the importance of each support vector.  $y_i$ : Denotes the class labels associated with the support vectors.  $K(x, x_i)$ : Is the kernel function, a crucial element allowing SVM s to operate effectively in high-dimensional spaces and capture non-linear relationships.  $b$ : Represents the bias term, allowing the SVM to offset the decision boundary from the origin.

#### 3.3.2 Decision Tree

DT inherently performs feature selection, is easy to interpret, can handle both numerical and categorical data, and is less prone to overfitting with the proper precautions

(Leo et al., 1984). They also work well with ensemble techniques, a key component of our methodology.

The data-driven decision-making process is significantly enhanced by the adaptability of DT models, enabling seamless preprocessing of nominal data and proficient handling of numerical and categorical variables (Myles et al., 2004). The algorithm's formulation involves parameters  $q$ , encompassing thresholds at each node and the chosen features for splits, represented as

$$h(x) = T(x, \theta) \quad (2)$$

Where  $h(x)$  denotes the output (prediction) for input features  $x$ , and  $T(x, \theta)$  signifies the DT model parameterized by  $\theta$ . After construction of the tree, nodes in the tree represent different transaction characteristics, and the thresholds are well chosen so as to help in picking situations where the possible transaction may be a fraud. For example, a node might evaluate whether some characteristics conform to prespecified lower thresholds and send transactions with these features to a branch associated with fraud.

### 3.3.3 Random Forest

RF is identified to be an effective supervised ML algorithm more suitable for regression and classification tasks for predicting fraudulent nature of transactions (Cutler et al., 2012; Ho, 1995). This elastic approach adequately deals with intervals for regression analysis as well as nominal for classification. RF on the other hand is a flexible method works as a committee of DTs and seeks for the better prediction (Khan et al., 2022).

According to Bagging (Bootstrap Aggregation), Random Forest (RF), each model is constructed by randomly choosing a subsample of the dataset with replacement, this procedure is called bootstrap (Mohapatra et al., 2022). Every model is trained individually, and the median of output of all the models is used to get the end projection. An RF predictor is calculated as follows:

$$y^c(x) = \frac{1}{C} \sum_{i=1}^c P_i(x) \quad (3)$$

Where  $y^c(x)$ : The probability or distribution of class  $c$  for the input features  $x$  unfamiliar you will get from the equation in outcome  $x$ .  $C$ : The total number of classes to which the items in the classification problem belong to.  $P_i(x)$ : The anticipated chance of the input features  $x$  belonging to the class  $i$ .

In particular, the impurity of every node in the DT construction is measured using Gini Importance. The average values of individual trees are part of the RF model's final feature significance phase. To do this, add up all of the feature importance values for each tree, then divide that amount by the number of trees. The following formula can be used to determine node importance using Gini Importance:

$$k_i = w_i G_i - w_{left(i)} G_{left(i)} - w_{right(i)} G_{right(i)} \quad (4)$$

Where  $k_i$ : The Gini Importance score for the specific node  $i$  in the DT.  $w_i$ : The weighted number of samples reaching node  $i$ .  $G_i$ : The impurity (Gini index) of node  $i$ , measuring the node's ability to classify samples correctly.  $w_{left(i)}$ : The weighted number of samples reaching the left child node of  $i$ .  $G_{left(i)}$ : The impurity of the left child node of  $i$ .  $w_{right(i)}$ : The weighted number of samples reaching the right child node of  $i$ .  $G_{right(i)}$ : The impurity of the right child node of  $i$ .

### 3.3.4 XGboost

The cutting-edge ML method known as XGBoost, or eXtreme Gradient Boosting, is well-known for its effectiveness and capacity for prediction. It is a member of the EL family, specifically boosting, and has become well-known because to its capacity to manage intricate data links and provide exceptional performance on a variety of tasks. By adding regularization strategies, parallel processing, and a scalable framework, XGBoost expands on the ideas of gradient boosting in this paper, making it a reliable option for identifying fraudulent transactions in credit card datasets.

XGBoost is excellent at identifying complex patterns in large feature sets thanks to its ensemble methodology. When working with unbalanced datasets, the regularization methods built into XGBoost provide robustness by reducing overfitting. Furthermore, XGBoost's adaptability in managing both linear and non-linear relationships fit very nicely with the subtleties of fraud detection, guaranteeing strong precision and recall metrics even in difficult situations.

$$\hat{y}(x) = \sum_{k=1}^K f_k(x) \quad (5)$$

where  $x$  stands for the features that were taken out of the transaction data and used as the XGBoost model's input. The ensemble's overall forecast is influenced by the total number of DTs, denoted by  $k$ . The output of each DT in the ensemble for the specified input features  $x$  is denoted by  $f_k(x)$ . These separate tree forecasts are combined to get the final prediction.

### 3.3.5 AdaBoost

Adaptive Boosting, or AdaBoost, is an EL technique that iteratively modifies the weights of weak models according to their prior errors in order to enhance their performance. In this study, AdaBoost is applied to detect fraudulent credit card transactions, where the aim is to enhance the accuracy of predictions by boosting the performance of simple base models, such as DT, through a series of iterations.

AdaBoost works by assigning higher weights to misclassified instances, encouraging subsequent weak learners to focus on these harder-to-predict cases. This process of reweighting and combining the weak models' outputs allows AdaBoost to create a strong model that performs well, especially when dealing with imbalanced datasets. Its ability to adapt to complex patterns and its sensitivity to the nuances of the data

make it highly effective for fraud detection tasks, particularly in scenarios with a high class imbalance between fraudulent and non-fraudulent transactions.

$$\hat{y}(x) = \sum_{k=1}^K \alpha_k f_k(x) \quad (6)$$

Where  $x$  refers to the features extracted from the transaction data, and  $f_k(x)$  represents the weak models (usually a DT) at iteration  $k$ .  $\alpha_k$  is the weight assigned to each weak models based on its performance during the training process. The final prediction is an aggregated output of all weak models, weighted according to their performance.

### 3.3.6 Multi-Layer Perceptron

Multilayer Perceptrons (MLPs) form the basic building block in the neural networks, which comprises layers of nodes interconnected, and possibility of analyzing high order correlations present within the statistics (Oukhouya & El Himdi, 2023). As categorized beneath the feedforward neural community, MLP contains an enter layer, one or multiple hidden layers, and an output layer. These models have proved diverse suitable for different applications and therefore are the basic selection for different types of learning machines (Koukaras et al., 2022). Remember that MLPs may efficiently capture the most intricate aspects of the patterns and dependencies found in the data when it comes to sequence data analysis, including stylometry features.

In this study, Multilayer Perceptrons (MLPs) are used due to its characteristic of search for setting of dependency/relationship existing in the transactional Structure. In this context, MLPs perform very well in identifying the fine-grained temporal patterns due to their capability of learning the relevant features inherent in the transactional data as a result of its hierarchical structure. This flexibility allows MLPs to accurately select fraudulent patterns at different levels of transaction aggregation, thus making them a suitable choice for fraud in economic transactions.

The forward pass of a standard MLP is characterized by follow Eq.

$$h^{(1)} = ReLU(W^{(1)}.x + b^{(1)}) \quad (7)$$

$$h^{(2)} = ReLU(W^{(2)}.h^{(1)} + b^{(2)}) \quad (8)$$

$$h^{(L)} = ReLU(W^{(L)}.h^{(L-1)} + b^{(L)}) \quad (9)$$

$$y = Softmax(W^{(L+1)}.h^{(L)} + b^{(L+1)}) \quad (10)$$

Here,  $x$  represents the transactional features inherent in the process and as such as inserted to the MLP.  $L$  Determines the complexity of the MLP structure, the more layers the model has, the higher the potential of the model to learn intricate patterns. ELEMENT Non linearity applied through ReLU to the result of the weighted inputs summed within each layer of hidden state.  $W^{(L)}$  and  $b^{(L)}$  are the learnable param-



eters, which indicates how much each input feature affect each layer.  $Y$  represents the output as determined by the model which when analyzed with the aid of the patterns learnt within the transactional data serves as alerts that may point to fraudulent transactions.

### 3.3.7 Artificial Neural Network (ANN)

A potent ML technique that draws motivation from the composition and operations of the human brain is the ANN (RB & KR, 2021). It is a useful tool for many tasks, including fraud detection, because it is especially good at capturing intricate, non-linear relationships within data. The ANN model's capacity to learn from high-dimensional and non-linear patterns is demonstrated in this work by analyzing the features taken from credit card transaction data.

ANNs are composed of layers of interconnected neurons, where each layer processes the input data and passes it on to the next. The model learns through backpropagation, adjusting weights to minimize prediction error. This iterative process allows the model to make highly accurate predictions based on the patterns observed during training. The flexibility of ANNs to adapt to various data distributions and their capacity to generalize well on unseen data make them effective for fraud detection, particularly in detecting subtle fraudulent patterns that may be difficult for simpler models to capture.

$$\hat{y}(x) = f(W.x + b) \quad (11)$$

Where  $x$  refers to the stylometry features derived from the document or transaction data,  $W$  represents the weight matrix of the network, and  $b$  is the bias term.  $f$  is the activation function used to process the weighted input and bias, generating the output prediction  $\hat{y}(x)$ .

## 3.4 Evaluation Metrics

In this work, only one train/test split approach is used. From our models, we evaluated their abilities using *Accuracy*, *Precision*, *Recall*, and *F1 Score*. These metrics can be used for classification problems only as they give more insight of the model's performance. These are computed from the confusion metrics as indicated in Table 8 below. These confusion metrics are defined below:

1. True Positives (TP) (Khalid et al., 2023), it is a scenario where detected fraud conducted by model is correct. In our case true positive is when the model gave a positive prediction and indeed it is a genuine fraud that has to be detected by the model. And this shows that the model correctly isolates or flag suspicious transactions as those that are fraudulent.

**Table 8** Confusion metrics

|        |   |           |    |
|--------|---|-----------|----|
| Actual | 0 | TN        | FP |
|        | 1 | FN        | TP |
|        |   | 0         | 1  |
|        |   | Predicted |    |

2. True Negatives (TN) (Mythili & Shanavas, 2014), means cases where the model is correct in their classification of non-fraudulent transactions. In the case of our study, true negative means that during the course of the study and using the model, a transaction which is classified as non-fraudulent (negative prediction) is indeed non-fraudulent (negative label). This demonstrates that the proposed model effectively isolates and predicts non-fraudulent transactions based on the features resultant from the related domain.

3. False Positives (FP) (Oma et al., 2019), recall that the Correct Rejections (CR) are cases in which the model does not label a transaction as fraudulent when in fact it is not, or false negatives (FN) while False Positives (FP) are cases where the model indicates a fraud transaction when it actually is not. False positive occurs when the model asses a transaction to be fraud, while in fact it is otherwise (Positive prediction: False Label Negative).

4. False Negatives (FN) (Alarfaj et al., 2022), represent instances where the model incorrectly picks a transaction as non-fraudulent whilst it's fraudulent. A false negative happens while the model identifies a transaction as non-fraudulent (negative prediction), however the real outcome exhibits it to be fraudulent (positive label). This suggests that the model failed to identify fraudulent activity when it occurred due to an inaccurate negative estimate.

Accuracy (Karkaba et al., 2023): Accuracy quantifies the percentage of all predictions that are accurate or the overall correctness of the predictions. It is calculated as follows:

$$\text{Accuracy (Acc)} : \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (12)$$

**Recall** (True Positive Rate) (Swathi et al., 2022): Recall, also known as sensitivity, measures the model's ability to correctly identify actual positives. It is calculated as:

$$\text{Recall} : \frac{(TP)}{(TP + FN)} \quad (13)$$

**Precision** (Ampomah et al., 2020): Precision quantifies the percentage of accurately predicted positive instances out of all cases the model anticipated to be positive, or the percentage of positive identifications that were actually correct. It is calculated as:

$$\text{Precision (Prec)} : \frac{(TP)}{(TP + FP)} \quad (14)$$

**F1 Score** (Ampomah et al., 2020): The F1 score is the harmonic mean of Precision and Recall, providing a balance between them. It is calculated as:

$$\text{F1 Score} : 2 \times \frac{\text{Precision} \times \text{Recall}}{(\text{Precision} + \text{recall})} \quad (15)$$

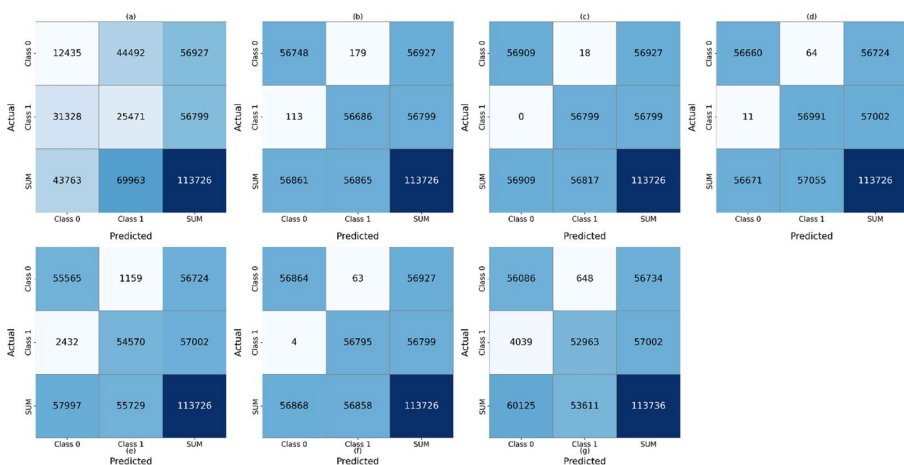
## 4 System Configuration

For testing the extent of the fraudulent characteristic of the transactions, an experimental design was made using a computer with Intel Core i7 processor and 8GB RAM. The primary implementation of the models was done in Scikit-learn (Liu, 2020), an open-source ML library in the Python language. Furthermore, to address our research objectives, Google Colab (Liu, 2020), an open-source web collaboration tool that supports reports containing live-code, interactive visualizations, equations, and constitutional text, was used, and to coordinate the construction of the reports hand in with collaborators. Throughout the study, Python served as the primary execution and evaluation language (Hafeez et al., 2021).

## 5 Results and Discussion

Each model's overall outcome is evaluated mainly using key metrics, as well as accuracy, precision, recall, and F1 score, with model validation performed through 10-fold cross-validation. Within this framework, Class 0 denotes transactions that are not fraudulent, whereas Class 1 denotes transactions that are fraudulent.

As shown in Fig. 2, both XGBoost and RF become the top performing models when their confusion matrices for Dataset 1 is plotted. With 11 false positives and 4 false negatives, XGBoost achieves near perfect precision and recall of 125 and 100, respectively, one of the key strengths of the method lies in its ability to capture complex feature interactions. Likewise, RF is seen to also perform well, with little misclassifications (false positive rate of 0.02% and false negative rate of 0.03%), due to its ensembling nature and ability to lower variance through bootstrap aggregation and feature randomness. Generalization capability of both models are very good and they are adequate for the size and complexity of the given dataset.

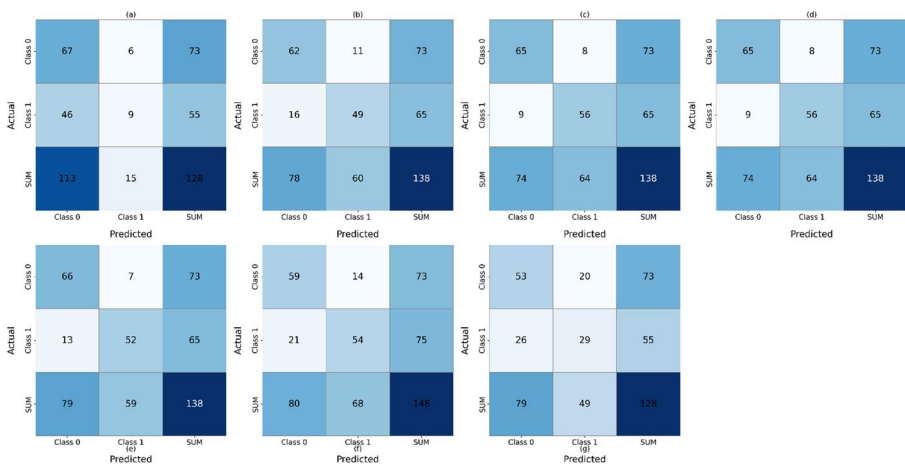


**Fig. 2** Confusion Metrics of Models on Dataset 1 (a) SVM (b) DT (c) RF (d) XGBoost (e) Adaboost (f) MLP (g) ANN

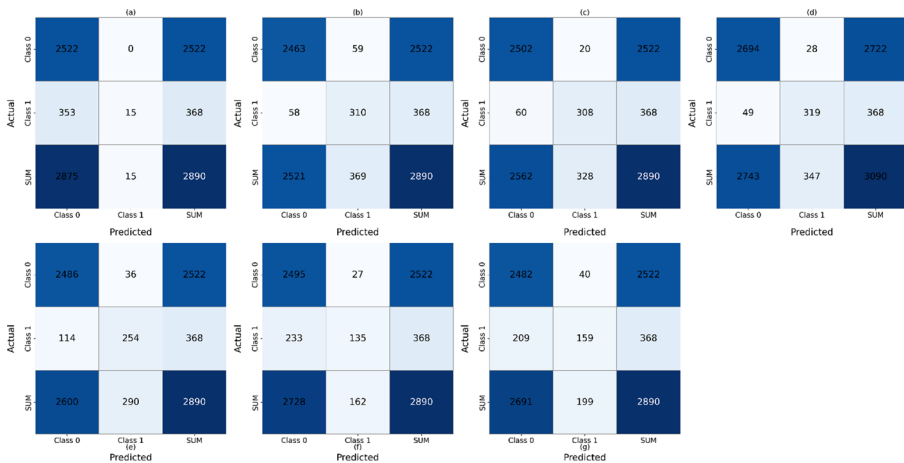
In contrast, SVM and DT are the lowest-performing models. SVM struggles with a significant false positive rate (39.12%) and poor recall for Class 1 (22.40%), which can be attributed to the dataset's high dimensionality and the difficulty of separating classes when features are heavily correlated or overlapping, challenging the kernel's ability to find an optimal hyperplane. DT performs slightly better but still has high misclassification rates (27.55% false negatives and 39.12% false positives), primarily due to its tendency to overfit on training data in high-variance scenarios, particularly when feature distributions are complex or noisy, even in balanced datasets, resulting in suboptimal generalization.

Similarly, in Dataset 2, the RF(c) model excelled with 87.68% accuracy, achieving high precision and recall for both Class 0 (non-fraudulent) and Class 1 (fraudulent), while the DT (b) also performed well with 80.43% accuracy, demonstrating the effectiveness of decision trees in handling non-linear patterns and diverse data attributes as shown in Fig. 3. In contrast, SVM (a) and Adaboost (e) underperformed, with SVM achieving only 29.23% recall for Class 1, indicating many false negatives, and Adaboost struggling to maintain stability, attaining only 31.88% recall for Class 1. These weaknesses stem from SVM's limitations in managing overlapping class distributions and Adaboost's sensitivity to outliers and noise.

From the confusion matrices for Dataset 3 (unbalanced), the RF (c) model emerged as the top performer, achieving 97.23% accuracy with strong precision and recall for both Class 0 (non-fraudulent) and Class 1 (fraudulent) as shown in Fig. 4. Its balanced performance in detecting fraudulent transactions, even in the presence of an imbalanced dataset, highlights its robustness, likely due to its ensemble nature that integrates multiple DTs. Similarly, XGBoost (d) and DT(b) also performed well, with accuracies of 97.34% and 95.95%, respectively, demonstrating strong adaptability to the imbalanced data. On the other hand, SVM (a) and Adaboost (e) depicted poor results. SVM's 4.08% recall for Class 1 indicates a significant number of false negatives, making it unsuitable for fraud detection in this case. Adaboost struggled with



**Fig. 3** Confusion Metrics of Models on Dataset 2 (a) SVM (b) DT (c) RF (d) XGBoost (e) Adaboost (f) MLP (g) ANN



**Fig. 4** Confusion Metrics of Models on Dataset 3 (a) SVM (b) DT (c) RF (d) XGBoost (e) Adaboost (f) MLP (g) ANN

**Table 9** Performance of the models across different metrics on dataset 1

| Models        | Accuracy | Precision | Recall | F1 Score |
|---------------|----------|-----------|--------|----------|
| <b>SVM</b>    | 0.33     | 0.2184    | 0.2841 | 0.2470   |
| Decision Tree | 0.99     | 0.99      | 0.99   | 0.99     |
| Random Forest | 0.99     | 0.99      | 0.99   | 0.99     |
| XG Boost      | 0.99     | 0.99      | 0.99   | 0.99     |
| Ada Boost     | 0.97     | 0.98      | 0.96   | 0.97     |
| MLP           | 0.99     | 0.99      | 0.99   | 0.99     |
| ANN           | 0.96     | 0.99      | 0.93   | 0.96     |

imbalanced data, achieving only 36.68% recall for Class 1, which compromises its ability to identify fraudulent transactions effectively. These results underscore the importance of using ensemble and tree-based models for handling imbalanced datasets in fraud detection (Fig. 4).

On Dataset 1, XGBoost, RF, MLP, and DT achieve near-perfect performance across accuracy, precision, recall, and F1 score (0.99), indicating their ability to handle the dataset effectively due to EL, boosting techniques, and complex decision boundaries as shown in Table 9. XGBoost, in particular, benefits from its gradient boosting framework, excelling in reducing errors and managing class imbalance. SVM, however, performs poorly with an accuracy of 0.33 and low F1 score (0.2470), attributed to its sensitivity to imbalanced data and difficulty in high-dimensional feature spaces, where the kernel struggles to separate classes optimally. ANN and AdaBoost show slightly lower metrics due to limitations in fully addressing the dataset's complexity and imbalance, though they still perform competitively compared to SVM.

On Dataset 2, RF and XGBoost are the top-performing models with accuracy and F1 scores of 0.88, benefiting from their ability to handle difficult feature interactions and reduce overfitting through ensemble methods as shown in Table 10. AdaBoost also performs well (accuracy 0.86, F1 score 0.84) due to its iterative focus on misclassified samples. The DT achieves moderate performance (accuracy 0.80), but

**Table 10** Performance of the models across different metrics on dataset 2

| Models               | Accuracy | Precision | Recall | F1 Score |
|----------------------|----------|-----------|--------|----------|
| <b>SVM</b>           | 0.62     | 0.76      | 0.29   | 0.42     |
| <b>Decision Tree</b> | 0.80     | 0.81      | 0.75   | 0.78     |
| <b>Random Forest</b> | 0.88     | 0.88      | 0.86   | 0.87     |
| <b>XG Boost</b>      | 0.88     | 0.88      | 0.86   | 0.87     |
| <b>Ada Boost</b>     | 0.86     | 0.88      | 0.80   | 0.84     |
| <b>MLP</b>           | 0.75     | 0.76      | 0.68   | 0.72     |
| <b>ANN</b>           | 0.67     | 0.66      | 0.60   | 0.63     |

**Table 11** Performance of the models across different metrics on dataset 3

| Models               | Accuracy | Precision | Recall | F1 Score |
|----------------------|----------|-----------|--------|----------|
| <b>SVM</b>           | 0.88     | 1.0       | 0.08   | 0.04     |
| <b>Decision Tree</b> | 0.96     | 0.84      | 0.84   | 0.84     |
| <b>Random Forest</b> | 0.97     | 0.94      | 0.84   | 0.89     |
| <b>XG Boost</b>      | 0.97     | 0.92      | 0.87   | 0.89     |
| <b>Ada Boost</b>     | 0.95     | 0.88      | 0.69   | 0.77     |
| <b>MLP</b>           | 0.91     | 0.83      | 0.37   | 0.51     |
| <b>ANN</b>           | 0.91     | 0.80      | 0.43   | 0.56     |

its single-tree structure limits generalization compared to ensemble models. SVM and ANN underperform with accuracy scores of 0.62 and 0.67, respectively, largely due to SVM's sensitivity to imbalanced data and ANN's requirement for extensive parameter tuning and more data for effective training. MLP, with accuracy 0.75, performs better than ANN but struggles with the dataset's complexity compared to ensemble models.

On Dataset 3 (imbalanced), RF and XGBoost models gave best results, both achieving an F1 score of 0.89, with high accuracy (0.97) as shown in Table 11. Their strength lies in their ability to handle imbalanced data effectively through EL and feature importance strategies. DT performs well (accuracy 0.96, F1 score 0.84) but lacks the robustness of ensemble models. AdaBoost (F1 score 0.77) struggles with imbalanced data as its iterative weighting can overemphasize minority class misclassifications. SVM (F1 score 0.04) significantly underperforms due to its sensitivity to class imbalance, focusing heavily on the majority class. Similarly, ANN and MLP perform moderately (F1 scores 0.56 and 0.51, respectively), likely due to suboptimal parameter tuning and their reliance on larger, balanced datasets for optimal performance.

It is clear from above results that the top-performing models, RF and XGBoost, excelled in both balanced and unbalanced datasets due to their robust EL techniques, which allow them to effectively handle difficult patterns and pick up the underlying links within the data. These models' ability to address class imbalances, especially with techniques like boosting (XGBoost) and bagging (RF), led to high precision, recall, and F1 scores across both datasets.

In contrast, models like SVM and Artificial Neural Networks (ANN) performed poorly on the unbalanced dataset, as their inherent sensitivity to class imbalances led to lower recall and F1 scores. SVM, with its focus on maximizing margins, struggled to properly classify minority fraudulent transactions, while ANN, despite its strong performance on balanced data, failed to generalize well to the imbalanced

**Table 12** Comparison with existing studies

| Ref                             | Year | Model                             | Accuracy | Precision | Recall | F1 Score |
|---------------------------------|------|-----------------------------------|----------|-----------|--------|----------|
| (Afriyie et al., 2023)          | 2023 | Random Forest                     | 0.96     | 0.09      | 0.97   | 0.17     |
| (Do et al., 2023)               | 2023 | XGBoost                           | 0.99     | 0.89      | 0.81   | 0.85     |
| (RB & KR, 2021)                 | 2021 | ANN                               | 0.99     | 0.81      | 0.76   | 0.93     |
| (Xuan et al., 2018)             | 2018 | Random Forest                     | 0.96     | 0.89      | 0.95   | 0.96     |
| This Study (Balanced dataset)   | 2024 | MLP, Random Forest, Decision Tree | 0.99     | 0.99      | 0.99   | 0.99     |
| This Study (Unbalanced dataset) | 2024 | Random Forest                     | 0.97     | 0.94      | 0.84   | 0.89     |

dataset, resulting in lower overall performance metrics. Our findings align with those of previous studies. For instance, (Bodepudi, 2021) reported that SVM achieved an accuracy of 70.01% in fraud detection, which was lower than that of Random Forest which achieved 99.74% accuracy. Similarly, (Musiliudeen et al., 2024) found that an SVM model, when applied to a severely imbalanced dataset, achieved an accuracy of 99.9%, but its precision was 82% and recall was only 32%, highlighting its poor performance in detecting fraudulent transactions. These studies further corroborate the limitations of SVM in handling imbalanced datasets and emphasize the need for more robust models in fraud detection tasks.

## 5.1 Comparative Analysis

The comparative analysis of the performance metrics across existing studies and our study reveals notable differences in the effectiveness of various ML models for fraudulent transaction detection. Afriyie et al. (2023) demonstrated moderate accuracy with a RF model, achieving 0.96, but yielded low precision at 0.09, indicating a high false positive rate. Similarly, Do et al. (Do et al., 2023) achieved higher accuracy with XGBoost at 0.99, but experienced lower recall and precision compared to our study. RB & KR, (2021) obtained commendable results with an ANN, particularly in terms of precision at 0.81, but our study surpassed these metrics across all performance measures. Xuan et al., (2018) showcased competitive performance with a RF model, yet our study consistently outperformed their results, especially in precision and F1 score. Notably, our study achieved superior performance across all evaluated metrics with the MLP, RF, and DT models on the balanced dataset, demonstrating exceptional accuracy, precision, recall, and F1 score of 0.99 across all measures. However, on the unbalanced dataset, while our RF model still achieved high performance with an accuracy of 0.97, it exhibited slightly reduced recall (0.84) and F1 score (0.89), highlighting the challenges of handling class imbalances. This differentiation underscores the importance of balanced datasets in maximizing the performance of ML models for fraud detection (Table 12).

## 6 Conclusion

It has been shown that ML techniques are very efficient for credit card transaction fraud event detection. This showed high accuracy of 0.99 with the balance set and through models including SVM, DT, RF, and Multilayer Perceptrons. But accuracy and recall of 0.97 and 0.84 on the unbalanced dataset indicates the difficulty of class imbalance in fraud recognition tasks.

This study provides implications for policy makers and financial institutes to develop more robust framework of anti fraud devices for use by financial institutes and regulatory bodies. Applying machine learning based solutions will help financial organizations to stay prepared and prevent various fraudulent activities, and reduce the financial losses of the organizations, at the same time, also building lesser misapprehension among their customers.

Nevertheless, there is still some room for improvement and desire for future directions. There is much future research that can be done to try and integrate DL architectures to pick up more difficult patterns in transactional data. Furthermore, it could also be coupled with real time monitoring and adaptive learning mechanisms which would allow the system to respond to the arising fraud tactics more swiftly. In addition, working with financial institutions, regulatory bodies and policymakers can identify ways in which to use financial institutions, regulatory bodies and policymakers to validate and refine the models in the real world. Overall, the machine learning and analytics are attracting further advances and the policy supports the banking and financial risks, it will likely enhance the fraud detection system and reducing the financial risks.

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